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2 Real Analysis 2023
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Test 1
 21 August 2023



Instructions:

- This question paper consists of **7 pages** (including this one).
- You have 90 minutes to answer the questions.
- Answer all questions in the spaces provided on this question paper. Use the backs of pages if needed.
- Use your *UCT Examination Answer Book* for rough work. The work in your UCT Examination Answer Book will not be marked. Only the answers that you write on this question paper will be marked.
- Give full proofs.
- Be careful to provide answers that we can read and make sense of at all times. If we can't read your handwriting, we will mark your solution incorrect.
- 50 points = 100% (52 points available).

Problem	1	2	3	4	Σ	Mark
Maximum Points	20	8	14	10	50	100%
Result						

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Problem 1

(20 points)

Prove or disprove.

- (a) (i) The sum of two rational numbers is a rational number.
- (ii) The sum of two irrational numbers is an irrational number.
- (b) (i) The union $A \cup B$ of two countable sets A, B is countable.
- (ii) The difference $A \setminus B$ of two countable sets A, B is countable.
- (iii) The difference $A \setminus B$ of two countable sets A, B is always finite.
- (c) $\bigcap_{n=1}^{\infty} \left[0, \frac{1}{\sqrt{n}}\right] = \{0\}$.
- (d) (i) The set $S = \left\{\frac{2n-1}{n} : n \in \mathbb{N}\right\}$ is bounded above.
- (ii) Let $S = \left\{\frac{2n-1}{n} : n \in \mathbb{N}\right\}$, then $\sup S = 1$.
- (iii) Let $S = \left\{\frac{2n-1}{n} : n \in \mathbb{N}\right\}$, then $\sup S = 2$.

Solution.

- (a) (i) True. Let $x, y \in \mathbb{Q}$, then $x = \frac{m}{n}$, $y = \frac{p}{q}$ with $m, n, p, q \in \mathbb{Z}$, $n, q \neq 0$. Then $x + y = \frac{mq+np}{nq}$, where $mq + np, nq \in \mathbb{Z}$, $nq \neq 0$. Thus, $x + y \in \mathbb{Q}$.
(2 points)
- (ii) False. For example, the numbers $x = \sqrt{2}$ and $y = -\sqrt{2}$ are irrational, but their sum $x + y = 0$ is rational.
(2 points)
- (b) (i) True. Since A and B are countable, they can be listed as sequences:
 $A = \{a_1, a_2, a_3, \dots\}$, $B = \{b_1, b_2, b_3, \dots\}$. Then we can write
 $A \cup B = \{a_1, b_1, a_2, b_2, a_3, b_3, \dots\}$ and delete the repeats.
(2 points)
- (ii) False. For example, the set $B = \{1, 2, 3, 5, 7, \dots\}$ is countable, but $B \setminus B = \emptyset$.
(2 points)
- (iii) False. For example, the sets $\mathbb{N}$ and B as above are countable, but their difference $\mathbb{N} \setminus B = \{2, 4, 6, 8, \dots\}$ is infinite (more precisely, countable).
(2 points)
- (c) True. We have $0 \in \left[0, \frac{1}{\sqrt{n}}\right]$ for every $n \in \mathbb{N}$, and consequently $0 \in \bigcap_{n=1}^{\infty} \left[0, \frac{1}{\sqrt{n}}\right]$. If $x < 0$, then x does not lie in any of the sets $\left[0, \frac{1}{\sqrt{n}}\right]$ and therefore $x \notin \bigcap_{n=1}^{\infty} \left[0, \frac{1}{\sqrt{n}}\right]$. If $x > 0$, then by the Archimedean Property there exists $n_0 \in \mathbb{N}$ with $0 < \frac{1}{n_0} < x^2$. The last inequality is equivalent to $\frac{1}{\sqrt{n_0}} < x$. Thus, $x \notin \left[0, \frac{1}{\sqrt{n_0}}\right]$ and hence $x \notin \bigcap_{n=1}^{\infty} \left[0, \frac{1}{\sqrt{n}}\right]$.

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An alternative proof of the fact that 0 is the only point in the intersection: Let $a \in \bigcap_{n=1}^{\infty} \left[0, \frac{1}{\sqrt{n}}\right]$. Then for any $n \in \mathbb{N}$ we have $a \in \left[0, \frac{1}{\sqrt{n}}\right]$. This implies $0 < a < \frac{1}{\sqrt{n}}$ for any $n \in \mathbb{N}$. Equivalently, $0 < a^2 < \frac{1}{n}$ for any $n \in \mathbb{N}$. By the Archimedean Property, this is only possible if $a^2 = 0$, i.e. $a = 0$.

(3 points)

- (d) (i) True. For any $n \in \mathbb{N}$ we have $\frac{2n-1}{n} \leq \frac{2n}{n} = 2$. Thus, 2 is an upper bound for S .

(2 points)

- (ii) False. For example, for $n = 2$ we have $\frac{2n-1}{n} = \frac{3}{2} > 1$, and thus 1 is not an upper bound for S .

(2 points)

- (iii) True. We have seen in (i) that 2 is an upper bound for S . Given $\varepsilon > 0$, we can find $n \in \mathbb{N}$ with $\frac{2n-1}{n} > 2 - \varepsilon$. This is because

$$\frac{2n-1}{n} > 2 - \varepsilon \iff 2 - \frac{2n-1}{n} < \varepsilon \iff \frac{2n - (2n-1)}{n} < \varepsilon \iff \frac{1}{n} < \varepsilon,$$

and we can find $n \in \mathbb{N}$ with $\frac{1}{n} < \varepsilon$ by the Archimedean Property.

(3 points)

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Problem 2

(8 points)

Let $(x_n)_{n=1}^\infty$ be the sequence defined recursively by

$$\begin{aligned} x_1 &= 7, & x_2 &= 27, \\ x_{n+2} &= 6x_{n+1} - 5x_n, & n &\in \mathbb{N}. \end{aligned}$$

Prove by induction that $x_n = 2 + 5^n$, $n \in \mathbb{N}$.

Solution.

The statement $P(n)$ in this case is “ $x_n = 2 + 5^n$ and $x_{n+1} = 2 + 5^{n+1}$ ”.
For the induction base, we consider $P(1)$. We have

$$x_1 = 2 + 5^1 = 7 \quad \text{and} \quad x_2 = 2 + 5^2 = 27.$$

Thus, $P(1)$ is true.

(2 points for the base)

For the induction step, we assume that $P(k)$ is true for some $k \in \mathbb{N}$, i.e. $x_k = 2 + 5^k$ and $x_{k+1} = 2 + 5^{k+1}$. To prove $P(k+1)$, we have to show that $x_{k+2} = 2 + 5^{k+2}$. We have

$$\begin{aligned} x_{k+2} &= 6x_{k+1} - 5x_k = 6(2 + 5^{k+1}) - 5(2 + 5^k) = 12 + 6 \cdot 5^{k+1} - 10 - 5 \cdot 5^k \\ &= 2 + 6 \cdot 5^{k+1} - 5^{k+1} = 2 + 5 \cdot 5^{k+1} = 2 + 5^{k+2} \end{aligned}$$

as desired.

(5 points for the step)

By the principle of the mathematical induction, $x_n = 2 + 5^n$ for all $n \in \mathbb{N}$.

(1 point for the understanding of induction)

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Problem 3

(14 points)

Read the following theorem and an outline of its proof.

Theorem. The set of real numbers $\mathbb{R}$ is uncountable.

Proof. Suppose that we can list the set of real numbers as a sequence $\mathbb{R} = \{x_1, x_2, x_3, \dots\}$. Choose a closed bounded nonempty interval $[a_1, b_1]$ such that $x_1 \notin [a_1, b_1]$. For each $n \in \mathbb{N}$, let $[a_{n+1}, b_{n+1}]$ be a subinterval of $[a_n, b_n]$ with $x_{n+1} \notin [a_{n+1}, b_{n+1}]$. Put $I = \bigcap_{n=1}^{\infty} [a_n, b_n]$. Then there exists $m \in \mathbb{N}$ with $x_m \in I$, which leads to a contradiction.

Answer the following questions:

- Define what is meant by a set A being uncountable.
- Explain how we know that $\mathbb{R}$ is not finite.
- Explain why we list $\mathbb{R}$ as a sequence in the beginning of the proof. What is our intention?
- Show how to choose an interval $[a_1, b_1]$ with $x_1 \notin [a_1, b_1]$.
- Explain how to find a subinterval $[a_{n+1}, b_{n+1}]$ of $[a_n, b_n]$ with $x_{n+1} \notin [a_{n+1}, b_{n+1}]$.
- Explain why there is $m \in \mathbb{N}$ with $x_m \in I$.
- How does the above lead to a contradiction?

Solution.

- A is uncountable when A is neither finite nor countable.
(2 points)
- We know that $\mathbb{N} \subset \mathbb{R}$ and $\mathbb{N}$ is countable (in particular, not finite). It follows that $\mathbb{R}$ is not finite.
(2 points)
- This is a proof by contradiction. We assume that $\mathbb{R}$ is countable, i.e. that its elements can be listed as a sequence.
(2 points)
- For example, we can take $[a_1, b_1] = [x_1 + 1, x_1 + 2]$.
(2 points)
- We divide $[a_n, b_n]$ into three thirds of equal lengths. At least one of these thirds does not contain x_{n+1} . We take a third with this property to be $[a_{n+1}, b_{n+1}]$.
(2 points)

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- (f) By the Nested Interval Property, $I = \bigcap_{n=1}^{\infty} [a_n, b_n] \neq \emptyset$, i.e., there is $x \in \mathbb{R}$ with $x \in I$. But all elements of $\mathbb{R}$ are listed in the sequence $\mathbb{R} = \{x_1, x_2, x_3, \dots\}$. Thus, $x = x_m$ with some $m \in \mathbb{N}$.

(2 points)

- (g) By construction of the intervals $[a_m, b_m]$ we know that $x_m \notin [a_m, b_m]$. It follows that $x_m \notin I = \bigcap_{n=1}^{\infty} [a_n, b_n]$. This contradicts the statement $x_m \in I$ from above.

(2 points)

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Problem 4

(10 points)

- (a) Let $(a_n)_{n=1}^{\infty}$ be a sequence in $\mathbb{R}$ and $a \in \mathbb{R}$. Define what it means that $\lim_{n \rightarrow \infty} a_n = a$.
- (b) Suppose that $\lim_{n \rightarrow \infty} a_n = 1$. Prove by definition that $\lim_{n \rightarrow \infty} a_n(a_n - 1) = 0$.

Solution.

- (a) $\lim_{n \rightarrow \infty} a_n = a$ means that for any given $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that $|a_n - a| < \varepsilon$ for all $n \geq N$, or, as a formula,

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N : |a_n - a| < \varepsilon.$$

(2 points)

- (b) By the definition of the limit $\lim_{n \rightarrow \infty} a_n = 1$, there exists $N_1 \in \mathbb{N}$ such that

$$|a_n - 1| < 1 \quad \text{for all } n \geq N_1.$$

The inequality $|a_n - 1| < 1$ is equivalent to $1 - 1 < a_n < 1 + 1$. We obtain that $|a_n| < 2$ for all $n \geq N_1$.

We have

$$|a_n(a_n - 1) - 0| = |a_n| |a_n - 1| < 2|a_n - 1| \quad \text{for all } n \geq N_1.$$

Given $\varepsilon > 0$, there exists $N_2 \in \mathbb{N}$ such that

$$|a_n - 1| < \frac{\varepsilon}{2} \quad \text{for all } n \geq N_2.$$

Put $N = \max\{N_1, N_2\}$. Then

$$|a_n(a_n - 1) - 0| < 2|a_n - 1| < 2 \cdot \frac{\varepsilon}{2} = \varepsilon \quad \text{for all } n \geq N.$$

This implies, by definition, that $\lim_{n \rightarrow \infty} a_n(a_n - 1) = 0$.

An alternative solution: The sequence $(a_n)_{n=1}^{\infty}$ is convergent. It follows that it is bounded. Thus, there is a number $M > 0$ such that $|a_n| \leq M$ for all $n \in \mathbb{N}$. We have

$$|a_n(a_n - 1) - 0| = |a_n| |a_n - 1| \leq M|a_n - 1|.$$

Given $\varepsilon > 0$, there exists $N_0 \in \mathbb{N}$ such that

$$|a_n - 1| < \frac{\varepsilon}{M} \quad \text{for all } n \geq N_0.$$

Then

$$|a_n(a_n - 1) - 0| \leq M|a_n - 1| < M \cdot \frac{\varepsilon}{M} = \varepsilon \quad \text{for all } n \geq N_0.$$

This implies, by definition, that $\lim_{n \rightarrow \infty} a_n(a_n - 1) = 0$.

(8 points)